Exercise 5. Determine for which natural number n the inequalities are satisfied simultaneously:

$$\binom{n}{8} > \binom{n}{7} \text{ and } \binom{n}{8} > \binom{n}{9}$$

Exercise 6. Find the number n given that: $\binom{n}{3} - \binom{n}{2} = 14$.

Exercise 7. Prove that:

$$\binom{3}{3} + \binom{4}{3} + \binom{5}{3} + \binom{6}{3} + \binom{7}{3} = \binom{8}{4}$$

Exercise 8. Prove that for any natural number $n \geq 1$, the number

$$\binom{2n}{n}$$

is divisible by $n + 1$ and by $4n - 2$